Week 45

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```
try:
    import fysisk_biokemi
    print("Already installed")
except ImportError:
    %pip install -q "fysisk_biokemi[pyodide] @ https://au-mbg.github.io/fysisk-biokemi/
wheels/fysisk_biokemi-0.2.2-py3-none-any.whl"
```

---

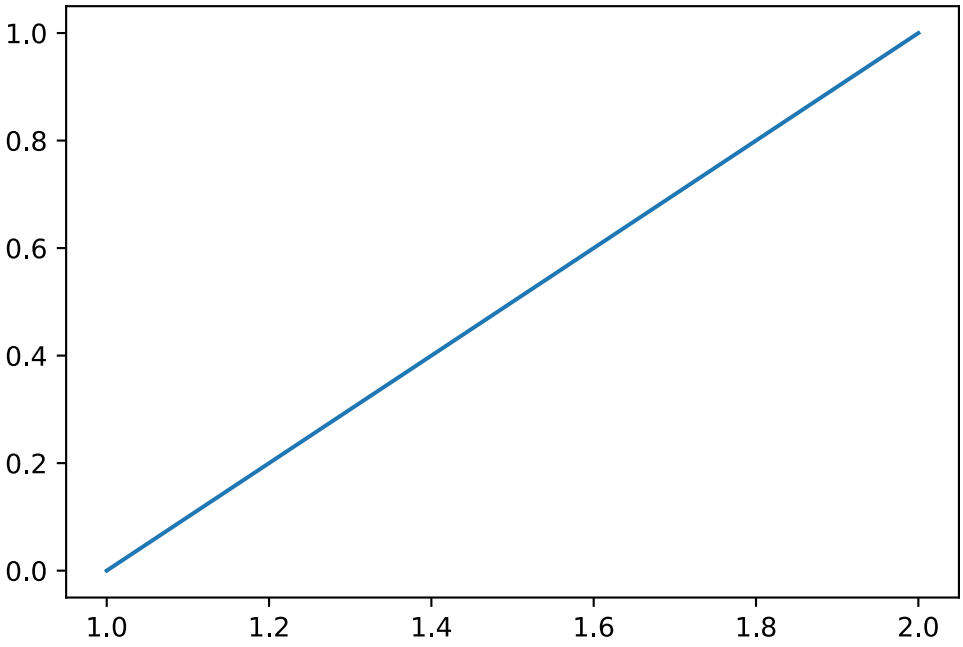
1 Python: Introduction to plotting

```
import numpy as np
import matplotlib.pyplot as plt
```

There are many ways to plot, we will be using the `matplotlib` library - but the concepts are similar across most ways of plotting.

We can make a plot of a straight line between two points (x_1, y_1) and (x_2, y_2) like so;

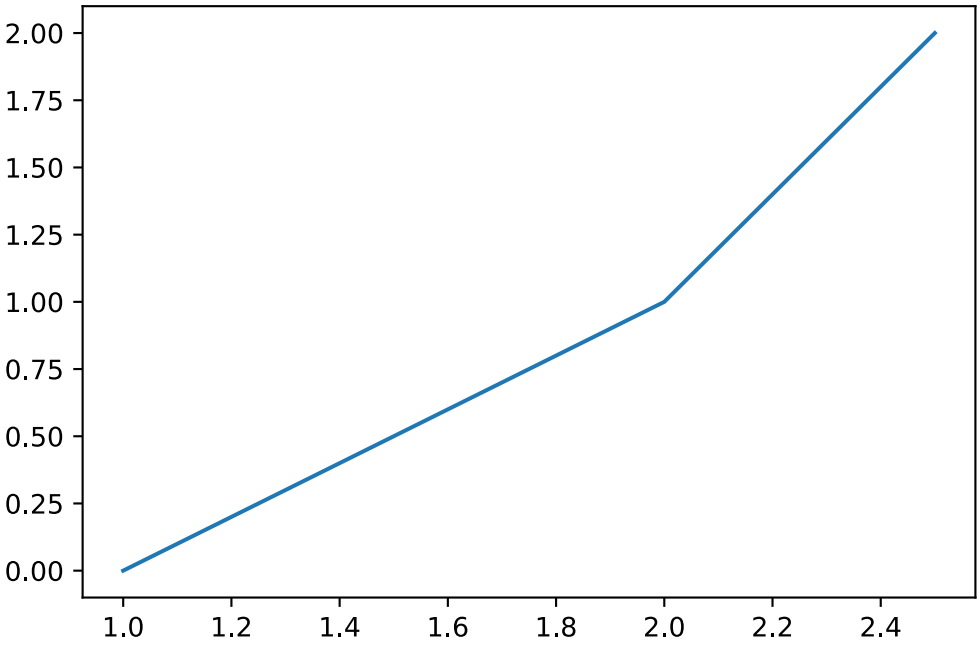
```
x = np.array([1, 2]) # This array contains all the x-coordinates
y = np.array([0, 1]) # And this contains all the y-coordinates.
fig, ax = plt.subplots(figsize=(6, 4))
ax.plot(x, y)
```



> **Note**
>
> `ax` is a special type (like `int`, `float`, `str`), you can think of it as the box that contains the plot.

If we want to connect to a third point (x_3, y_3) we would instead write

```
x = np.array([1, 2, 2.5])
y = np.array([0, 1, 2])
fig, ax = plt.subplots(figsize=(6, 4))
ax.plot(x, y)
```



Notice that with three sets of points we plot **two** line segments.

(a) Trying to plot a square

Make a plot with lines between the following 4

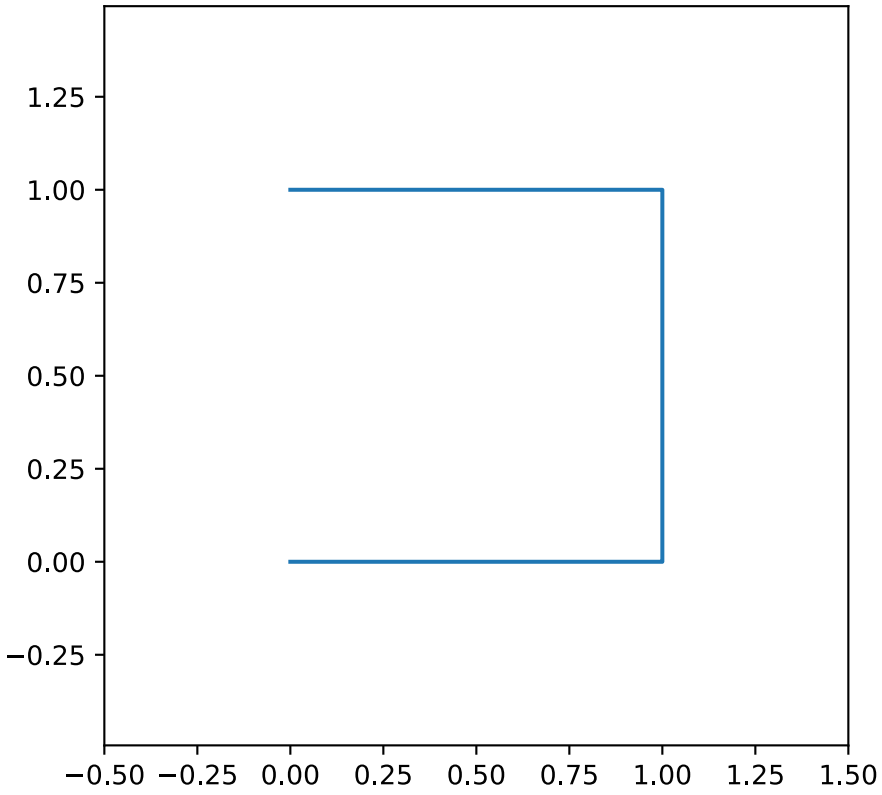
- $P_1 = (0, 0)$
- $P_2 = (1, 0)$
- $P_3 = (1, 1)$
- $P_4 = (0, 1)$

Start by defining an array for both `x` and `y`

```
x = np.array([0, 1, 1, 0])
y = np.array([0, 0, 1, 1])
```

Now use `ax.plot` to make the plot

```
fig, ax = plt.subplots(figsize=(5, 5))
ax.plot(x, y)
ax.axis('equal')
ax.set_xlim([-0.5, 1.5])
ax.set_ylim([-0.5, 1.5])
```



Consider the following;

- Why is it not a square?
- How many points did you use?
- How many line segments were plotted?

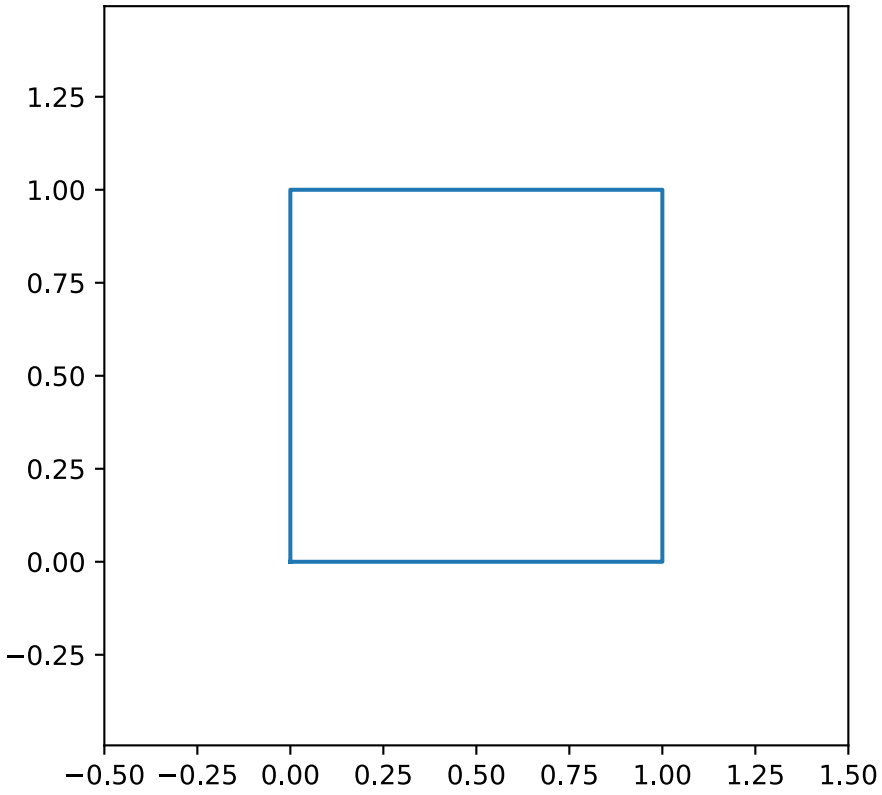
(b) Actually plotting a square

In order to actually plot a square you will need to update `x` and `y`, they both need to contain five numbers such that four line segments are plotted.

```
x = np.array([0, 1, 1, 0, 0])
y = np.array([0, 0, 1, 1, 0])
```

Copy your code for making the plot from the previous exercise to the cell below

```
fig, ax = plt.subplots(figsize=(5, 5))
ax.plot(x, y)
ax.axis('equal')
ax.set_xlim([-0.5, 1.5])
ax.set_ylim([-0.5, 1.5])
```



(c) Plotting a function

Often the data we plot can broadly be thought of as originating from a function, that is we have

$$y = f(x)$$

Where f is the function, this might for example be an experiment that produces some output y given some input x or it might be a traditional mathematical function like $y = x^2$. Plotting this type of data is exactly the same as plotting a square - it just usually consists of many more data points resulting in many line segments producing a smooth looking curve.

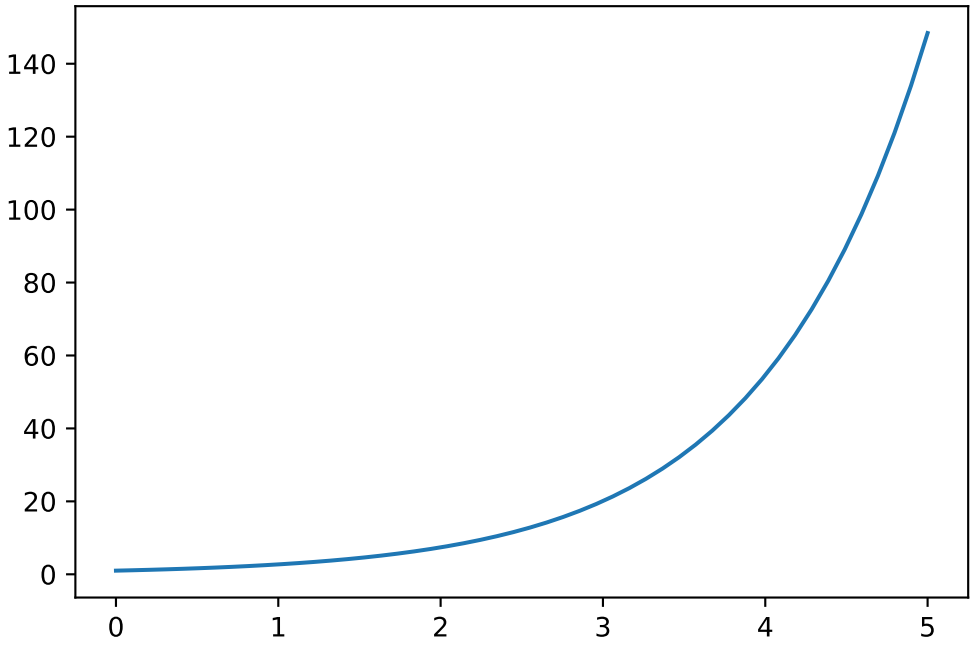
Make a plot of the function

$$y = \mathrm{e}^x$$

> **Note**
>
> The exponential function can be calculated using `np.exp` - likewise with the logarithm `np.log`.

```
x = np.linspace(0, 5, 50) # Make 50 points uniformly between 0 and 5.
y = np.exp(x) # Calculate y from x.

fig, ax = plt.subplots(figsize=(6, 4))
ax.plot(x, y)
```



Once you've made the plot try changing the number of points in `x` by changing the last argument in the call to `np.linspace`.

- What happens if you have very few points?

> **Tip**
>
> Remember that `np.exp` is used to calculate the exponential function.

(d) Plotting multiple functions

Often we want to plot multiple functions in the same figure as it enables us to compare them. Luckily, this is quite simple!

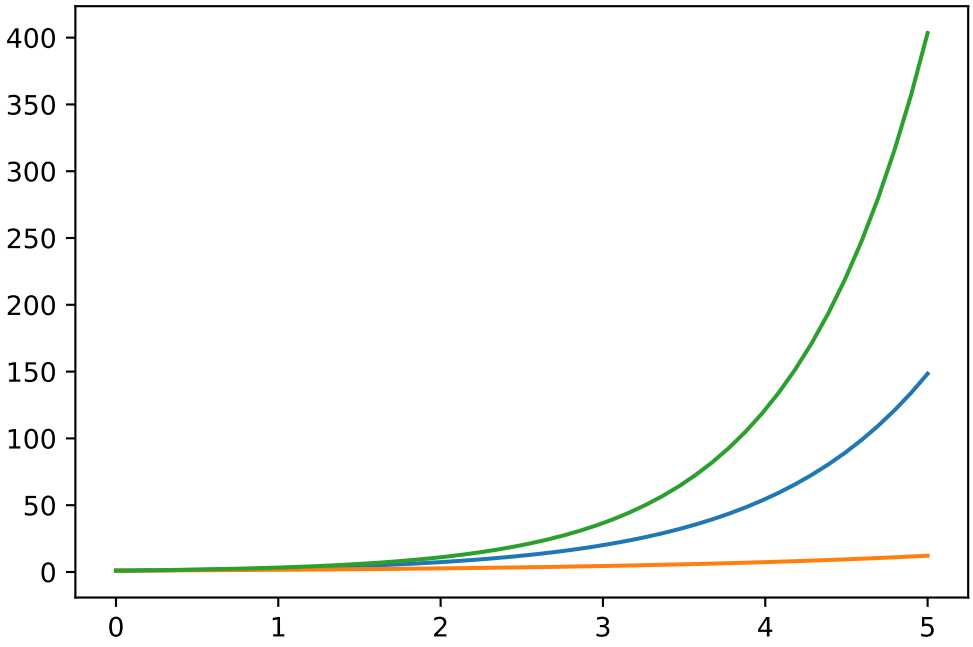
Plot the function

$$y = \mathrm{e}^{ax}$$

With $a = \left[1, \cfrac{1}{2}, 1.2\right]$ in the same plot producing three curves.

```
x = np.linspace(0, 5, 50) # Make 50 points uniformly between 0 and 5.
y1 = np.exp(1*x)
y2 = np.exp(0.5*x)
y3 = np.exp(1.2*x)

fig, ax = plt.subplots(figsize=(6, 4))
ax.plot(x, y1)
ax.plot(x, y2)
ax.plot(x, y3)
```



(e) Customizing plots

We will generally need to customize plots a little bit, for example the plot from the previous exercise doesn't have a label for the axis, there's no way of telling which curve is which and it has no title.

To add labels to the x and y axis we use the two functions

- `ax.set_xlabel('String with the name of the x-axis')`
- `ax.set_ylabel('String with the name of the y-axis')`

Adding information about the curves we plot can be done by also giving them a label, that is done as an argument to the `ax.plot` call, like so

```
fig, ax = plt.subplots()

ax.plot(x, y, label='Information about the curve.')
ax.legend()
ax.set_title('Title of plot')
```

Line 3

The `label` argument is given when the curve is plotted.

Line 4

`ax.legend()` is called to tell matplotlib to show the a 'legend' containing the `label` of all the plots that have been given one.

Line 5

Sets the title of the plot

Copy your code from the previous exercise and add labels for both the x and y axis aswell as each curve, for example you can label them according to the value of the the parameter a like `a = 1` etc.

```
... # Put some code here and try customizing it.
```

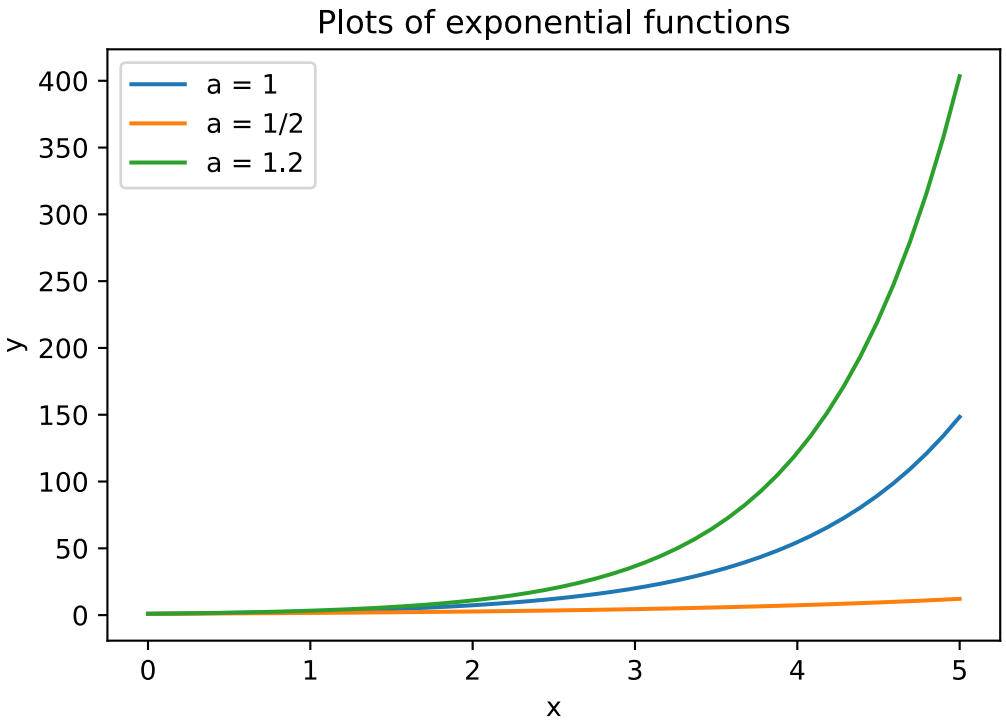
```
Ellipsis
```

```
x = np.linspace(0, 5, 50) # Make 50 points uniformly between 0 and 5.
y1 = np.exp(1*x)
y2 = np.exp(0.5*x)
y3 = np.exp(1.2*x)

fig, ax = plt.subplots(figsize=(6, 4))
ax.plot(x, y1, label='a = 1')
ax.plot(x, y2, label='a = 1/2')
ax.plot(x, y3, label='a = 1.2')

# Customization:
ax.set_xlabel('x')
ax.set_ylabel('y')
ax.legend()
ax.set_title('Plots of exponential functions')
```

```
Text(0.5, 1.0, 'Plots of exponential functions')
```



> **Note**
>
> There are many other customization options that can be added in the same way as the `label`, that is
>
> ```
> ax.plot(x, y, option_name=option_value)
> ```
>
> Some useful examples are
>
> - `linestyle`: Controls if the plot is made using a full, dashed or dotted line using `-`, `--` and `:` respectively. So `ax.plot(..., ..., linestyle = ":")` produces a dotted line.
> - `color`: Controls the color of the line - valid options are listed here https://matplotlib.org/stable/gallery/color/named_colors.html
> - `linewidth`: Sets the width of the line - can be any number.
> - `alpha`: Controls the transparency of the plot `0` being fully transparent and `1` being fully visible.
>
> You can try some of these options for the plot you've made above if you want to. There are *many* other ways of customizing plots for different situations. The matplotlib gallery shows a number of them.

(f) Other types of plots: Scatter

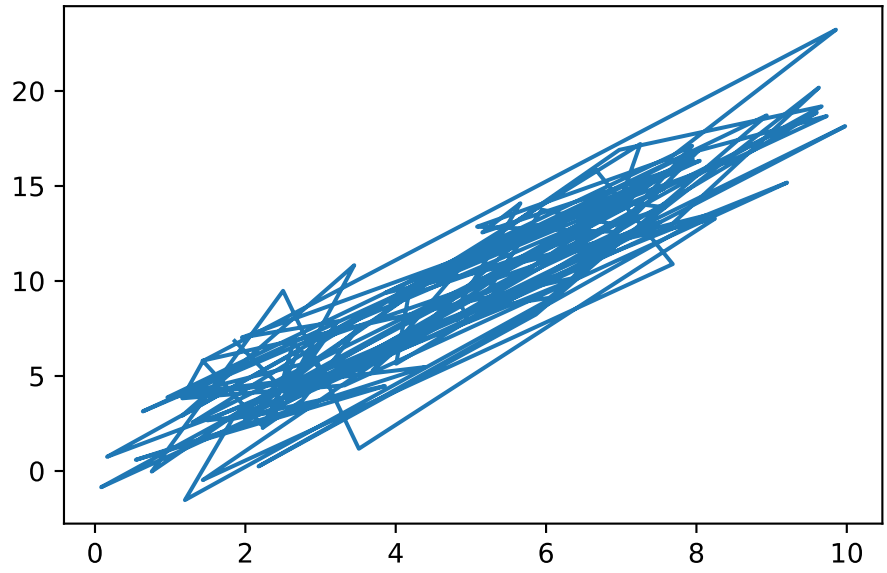
Sometimes we don't want to connect each point with a line segment but just show the points in a *scatter* plot.

The next cell makes some data

```
x = np.random.uniform(low=0, high=10, size=100)
y = 2*x + np.random.normal(scale=2, size=100)
```

Try plotting it like for the previous exercises

```
fig, ax = plt.subplots()
ax.plot(x, y)
```



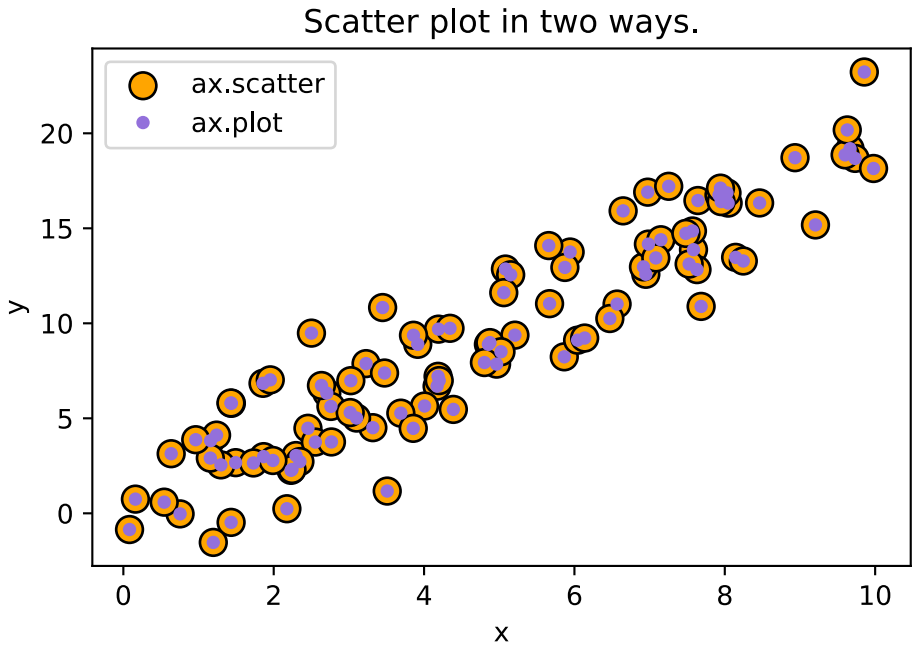
You will see a very strange plot like when you scribble out a word on a piece of paper. Clearly a line plot is not a very good way of showing this data! There are, at least, two ways of making a scatter plot

- `ax.plot(x, y, 'o')`: Quick and dirty way of just showing the points and not the lines.
- `ax.scatter(x, y)`: Function specifically for making these types of plots - with its own set of customization arguments.

Try either, or both, of these ways in the cell below

```
fig, ax = plt.subplots()

ax.scatter(x, y, s=100, label='ax.scatter', facecolor='orange', edgecolor='black',
linewidth=1)
ax.plot(x, y, 'o', label='ax.plot', color='mediumpurple', markersize=4)
ax.legend()
ax.set_xlabel('x')
ax.set_ylabel('y')
ax.set_title('Scatter plot in two ways.')
plt.show()
```



2 Average properties of amino acids and proteins.

```
import pandas as pd
import numpy as np
from IPython.display import display
```

In the accompanying Excel file (`averag-prope-amino-acids.xlsx`), you will find a table that contains the molecular weight of the 20 common amino acid residues, i.e. their weight as residues in a peptide chain. Additionally, you will find their relative frequency in E. coli proteins, where a frequency of 0.01 means that this residue constitutes 1 % of the residues in a protein.

(a) Load the data file.

Load the `averag-prope-amino-acids.xlsx` dataset

Use the cell below to load and display the dataset.

Need help loading the data?

The correct pandas function is already provided. You only need to set `data_file_path`:

1. Download the file named above from the Datasets page.
2. Upload the file to JupyterLite.
3. Replace `...` with the file's path in quotation marks. See Working with file paths for examples.

```
from IPython.display import display
from fysisk_biokemi.datasets import load_dataset
```

```
df = load_dataset('AA_frequency')
display(df)
```

	Name	MW of AA residue	Frequency in proteins
0	Alanine (Ala/A)	71.08	0.074
1	Arginine (Arg/R)	156.19	0.042
2	Asparagine (Asn/N)	114.11	0.044
3	Aspartic acid (Asp/D)	115.09	0.059
4	Cysteine (Cys/C)	103.15	0.033
5	Glutamic acid (Glu/E)	129.12	0.058
6	Glutamine (Gln/Q)	128.13	0.037
7	Glycine (Gly/G)	57.05	0.074
8	Histidine (His/H)	137.14	0.029
9	Isoleucine (Ile/I)	113.16	0.038
10	Leucine (Leu/L)	113.16	0.076
11	Lysine (Lys/K)	128.18	0.072
12	Methionine (Met/M)	131.20	0.018
13	Phenylalanine (Phe/F)	147.18	0.040
14	Proline (Pro/P)	97.12	0.050
15	Serine (Ser/S)	87.08	0.081
16	Threonine (Thr/T)	101.11	0.062
17	Tryptophan (Trp/W)	186.22	0.013
18	Tyrosine (Tyr/Y)	163.18	0.033
19	Valine (Val/V)	99.13	0.068

(b) Average molecular weight

Calculate the average molecular weight of a residue in a protein. To do this our procedure will be as follows

1. First calculate the residue weight times its frequency in proteins for each residue.
2. Sum these together to get the average molecular weight.

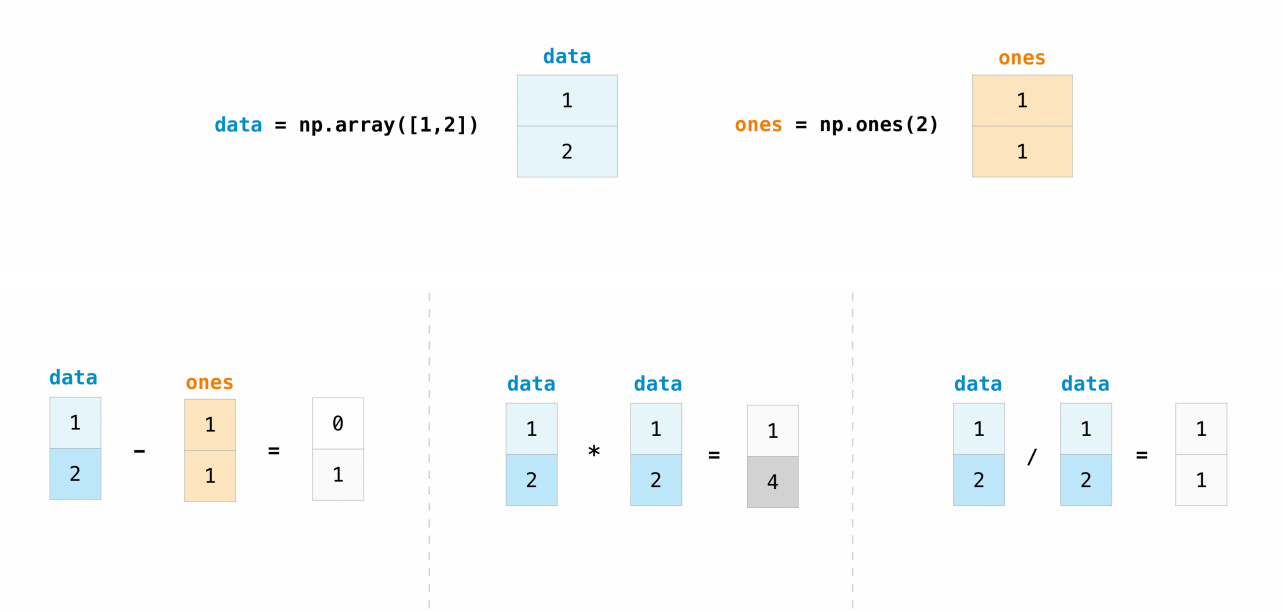
In the cell below finish the calculation of `weight_times_freq` by extracting the `"MW of AA residue"`-column and `"Frequency in proteins"` and multiplying them together.

Tip

You can index in the dataframe by using the column name, for example to get the `"MW of AA residue"`-column you would do

```
col = df["MW of AA residue"]
```

Arrays also allows us to operate on every element in the array at the same time, so arrays can be added, subtracted, multiplied, divided, etc, see for example this figure




```
weight_times_freq = df['MW of AA residue'] * df['Frequency in proteins']
```

Tip

You can use `np.sum` to sum all values in an array. For example

```
array = np.array([1, 2, 3, 4, 5])
sum_of_array = np.sum(array) # Gives 15
```

```
average_mw = np.sum(weight_times_freq)
unit_average_mw = "Da or g/mol"
print(f"{average_mw = :.3f} {unit_average_mw}")
```

```
average_mw = 110.566 Da or g/mol
```

> **Note**
>
> The syntax `f"{average_mw = :.3f}"` is just a way of printing the value with a nicer format, in this case we print the value to 3 decimal places. In Python these are called f-strings, you don't need to understand the details at the moment.

(c) Weight of 300-residue protein

What would the molecular weight of a 300-residue protein most likely be, if you did not know its sequence?

```
mw_300 = average_mw * 300
unit_mw_300 = "g/mol"
print(f"{mw_300 = :.3f} {unit_mw_300}")
```

```
mw_300 = 33169.941 g/mol
```

In many projects, you will be working with a mixture of proteins. This could for example be a cell lysate or a biological fluid for protein abundance analysis, or the early stages of a protein purification process. In these situations, you cannot work with a molecule specific extinction coefficient. Instead, we would use the average values, which we will determine below.

(d) Average concentration of residues

In many cases, we work with mixtures of proteins that do not have a defined molar extinctio coefficient. Instead, we work with an average 'mass extinction coefficient', which we will deduce here.

Calculate the average concentration of amino acid residues in a protein mixture at 1 mg/mL.

```
c_residue_avg = 1 / average_mw
unit_c_residue_avg = "M"
print(f"{c_residue_avg = :3.3f} {unit_c_residue_avg}")
```

```
c_residue_avg = 0.009 M
```

(e) Absorbance of an average mixture of residues

Calculate the absorbance from such a mixture under the assumption that only Trp and Tyr contribute, and that the path length is 1 cm.

```
freq = df.set_index("Name")["Frequency in proteins"]
f_trp = freq["Tryptophan (Trp/W)"]
f_tyr = freq["Tyrosine (Tyr/Y)"]

c_trp = c_residue_avg * f_trp
c_tyr = c_residue_avg * f_tyr
unit_c_trp_tyr = "M"
```

```
print(f"{c_trp = :3.5f} {unit_c_trp_tyr}")
print(f"{c_tyr = :3.5f} {unit_c_trp_tyr}")
```

```
c_trp = 0.00012 M
c_tyr = 0.00030 M
```

```
L = 1 # Path length 1 cm
A280_1mg_pr_ml = L * (5500 * c_trp + 1490 * c_tyr)
unit_A280_1mg_pr_ml = "(no unit)"
print(f"{A280_1mg_pr_ml = :3.3f} {unit_A280_1mg_pr_ml}")
```

```
A280_1mg_pr_ml = 1.091 (no unit)
```

In the compendium, we mentioned a "rule of thumb" that 1 mg/mL protein at a path length of 1 cm has an A280 = 1. How well does this compare to your calculated value?

(f) Absorbance → concentration

For a cell lysate, you measure and absorbance of 0.78 at a path length of 0.5 cm. What is the protein concentration?

```
# Set known values:
A = 0.78 # Unitless
l = 0.5 # cm

# Calculate concentration:
mass_conc = A / (A280_1mg_pr_ml * l)
unit_mass_conc = "mg/mL"
print(f"Protein concentration = {mass_conc:.3f} {unit_mass_conc}")
```

```
Protein concentration = 1.429 mg/mL
```

---

3 UV spectrum of apo- and holo-myoglobin

```
import matplotlib.pyplot as plt
import pandas as pd
from IPython.display import display
```

In this exercise we will analyze the spectra of apo- and holo-myoglobin. The dataset is given in `uv-spec-apo-holo-myo.csv`.

(a) Load the dataset

Load the dataset with pandas. Set `data_file_path` to the path of the dataset file named above.

Need help loading the data?

The correct pandas function is already provided. You only need to set `data_file_path`:

1. Download the file named above from the Datasets page.
2. Upload the file to JupyterLite.
3. Replace `...` with the file's path in quotation marks. See Working with file paths for examples.

```
from fysisk_biokemi.datasets import load_dataset
from IPython.display import display
df = load_dataset('apo_holo') # Load from package for the solution so it doesn't require to
interact.
display(df)
```

	wavelength	holo_absorbance	apo_absorbance
0	240	0.863073	0.455787
1	241	0.679118	0.404028
2	242	0.575443	0.365380
3	243	0.520933	0.337692
4	244	0.486763	0.318813
...	...	...	...
246	486	0.069077	0.033136
247	487	0.068888	0.033136
248	488	0.068391	0.033136
249	489	0.067608	0.033136
250	490	0.066691	0.033136

(b) Plot the spectra

Tip

To plot we use the `matplotlib` package. Plots are generally just straight lines connecting points, with enough points we get a smooth looking figure.

For example, to plot a line connecting three datapoints

```
fig, ax = plt.subplots()

x = np.array([1, 2, 3])
y = np.array([1, 4, 9])
ax.plot(x, y)
plt.show()
```

Line 1

This creates a `fig` and an `ax`, the `ax`-object is the box where our plot is created.

Line 3

Define the data to put on the x-axis.

Line 5

Does the actual plotting

Line 6

Shows the figure

There are *many* ways of customizing plots, you will see different ones in the exercises, but by no means all of them - if you are interested you can find more information on the matplotlib documentation.

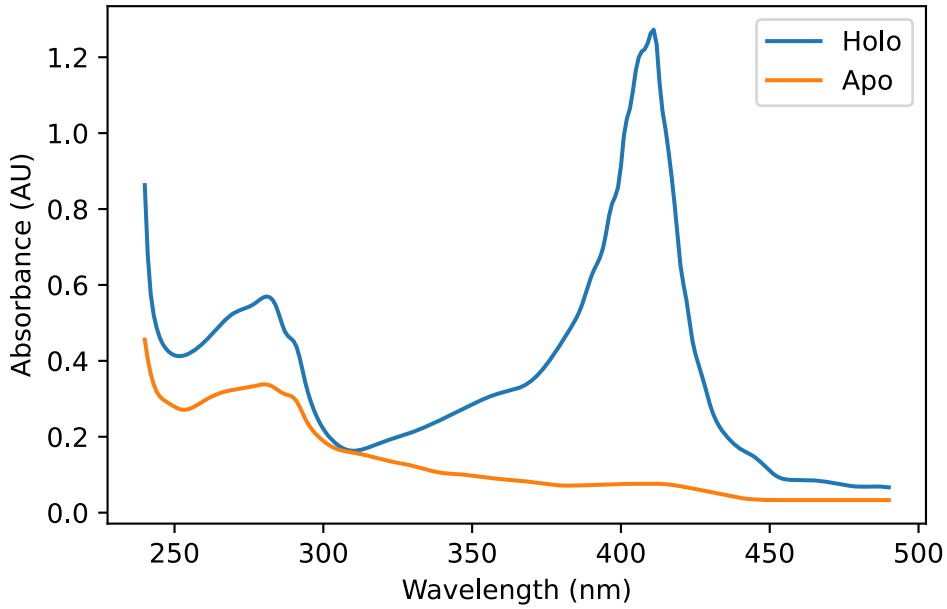
You don't have to worry about the `NaN` values in the dataset when plotting, matplotlib just skips plotting that line segment.

Using `matplotlib` plot each spectrum in the same figure as line plots.

```
fig, ax = plt.subplots()

ax.plot(df['wavelength'], df['holo_absorbance'], label='Holo')
ax.plot(df['wavelength'], df['apo_absorbance'], label='Apo')

ax.set_ylabel('Absorbance (AU)')
ax.set_xlabel('Wavelength (nm)')
ax.legend()
```



(c) Apo & Holo

Based on the spectra explain what ApoMb and HoloMb represent?

```
"""
ApoMb (Apo-myoglobin): Myoglobin without its heme prosthetic group.
"Apo" means the protein portion lacking its cofactor.

HoloMb (Holo-myoglobin): Complete myoglobin with the heme group bound.
"Holo" means the whole, functional protein with its cofactor.

The spectra show that HoloMb has strong visible light absorption due to
the heme group, while ApoMb lacks this characteristic absorption.
"""
```

You have learned that pure proteins without any UV/Vis-absorbing prosthetic groups bound, have basically no absorbance at wavelengths above λ>320 nm. Nevertheless ApoMb still show some absorbance above 320 nm. In this case, it can explained by the fact that ApoMb was generated from HoloMb by a procedure that will not be explained here.

(d) Absorption above 320 nm.

Explain why ApoMb in the spectrum above absorbs light at λ>320 nm.

```
"""
As ApoMb was generated from HoloMb, the ApoMb sample might still be contaminated
with some HoloMb
"""
```

(e) Efficiency estimate.

Give a rough estimate of the efficiency of the chosen procedure of heme-group removal. You may wish to look at the numerical values in the `DataFrame`.

```
"""
Compare the percentage by which the absorbance decreases at 410 nm ~94% removal
efficiency.
"""
```

(f) Isobestic points

Are there any isobestic points between the two spectra?

```
"""
Yes at around 300-310 nm.
"""
```

To get a better understanding of the causes for the different spectra, you can compare to litterature. The figure below shows the absorbance spectra of three states of myoglobin.

$$\mathrm{MbFe(II)O_2} \Rightarrow \mathrm{oxyMb}$$

$$\mathrm{MbFe(III)} \Rightarrow \mathrm{metMb}$$

$$\mathrm{MbFe(II)} \Rightarrow \mathrm{deoxyMb}$$

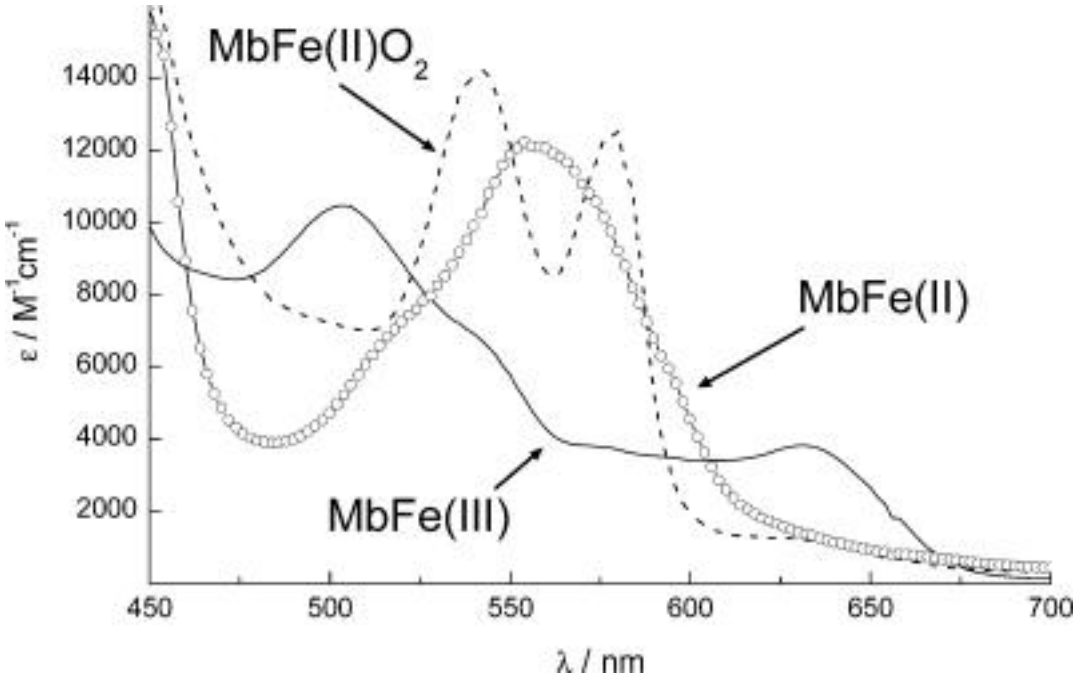


Figure 1: Absorbance spectra of different states of myoglobin. $\mathrm{MbFe(II)O_2}$ is "oxyMb", $\mathrm{MbFe(III)}$ is "metMb" and $\mathrm{MbFe(II)}$ is "deoxyMb". Taken from Barham et al (2010); ChemRev; 110 (4); pp2313-2365

(g) Aged myoglobin

metMb (MbFe(III)) is normally described as 'aged' myoglobin. What does this mean in terms of the bound iron?

```
"""In the aged state, the oxidation number of the iron is +3. The change in
oxidation number of iron (+2 vs +3) in fresh vs aged myoglobin means that the
electron configuration in the d- orbitals are different, which will affect the
way iron is bound in the heme group"""
```

(h) Change in absorbance

Give a qualitative explanation to the observed change in absorbance of metMb compared to fresh oxy-/deoxyMb?

```
"""
The changed in electronic configuration and the difference this causes to how
iron is bound in the hemegroup influences the absorption.Moreover binding of
oxygen via the iron again changes the electron configuration of the iron atom,
resulting in differences between deoxy- and oxymyoglobin
"""
```

(i) Type of shift

Is the spectral difference from deoxyMb to metMb a redshift or a blueshift?

```
"""
The absorbance peak in metMb is at lower wavelength (higher energy) as compared
to deoxyMb, and is therefore described as a blueshift.
"""
```

(j) Experiment design

You would like to set up an experiment where the absorbance of myoglobin at defined wavelength(s) should be used to measure the level of O_2 binding. Sketch how the absorbance spectra would look like when going from deoxyMb and continuously increasing the concentration of O_2 (draw this for five different O_2 concentrations going from pure deoxyMb to pure oxyMb)

---

4 Proteins in blood plasma.

```
import pandas as pd
from IPython.display import display
```

In this exercise we will learn how Python is excellent for handling datasets with many data points and how it can be used to apply the same procedure to all the data points at once.

A researcher wants to determine the concentration of two proteins in blood plasma that is suspected to be involved in development of an autoimmune disease. 500 patients and 500 healthy individuals were included in the study and absorbance measurements of the two purified proteins from all blood plasma samples were measured at 280 nm. The molecular weight and extinction coefficients of the two proteins are given in the table below.

Protein	M_w [kDa]	ϵ [$\mathrm{M^{-1}cm^{-1}}$]
1	130	180000
2	57	80000

(a) Load the dataset

Load the dataset as a dataframe from the file `protei-blood-plasma.xlsx` with pandas. Set `data_file_path` to the path of the dataset file.

> Need help loading the data?
>
> The correct pandas function is already provided. You only need to set `data_file_path`:
>
> 1. Download the file named above from the Datasets page.
> 2. Upload the file to JupyterLite.
> 3. Replace `...` with the file's path in quotation marks. See Working with file paths for examples.

```
from fysisk_biokemi.datasets import load_dataset
from IPython.display import display
df = load_dataset('protein_blood_plasma') # Load from package for the solution so it
doesn't require to interact.
display(df)
```

	A280_protein1_healthy	A280_protein1_patient	A280_protein2_healthy	A280_protein2_patient
0	0.086241	0.014437	0.035768	0.034756
1	0.095333	0.014813	0.040662	0.040242
2	0.091058	0.014119	0.039344	0.032946
3	0.105004	0.014430	0.041232	0.032568
4	0.097421	0.014083	0.035860	0.036829
...	...	...	...	...
495	0.090870	0.015723	0.031065	0.031464
496	0.084912	0.014091	0.036573	0.037934
497	0.097572	0.014300	0.043926	0.046982
498	0.096186	0.014589	0.040299	0.036045
499	0.079806	0.013657	0.030360	0.041318

(b) Calculate concentrations

Calculate the molar concentration of the two proteins in all samples, the light path for every measurement is 0.1 cm.

Always a good idea to assign known values to variables

```
protein_1_ext_coeff = 180000
protein_2_ext_coeff = 80000
l = 0.1
```

> **Tip**
>
> You can set new columns in a `DataFrame` by just assigning to it
>
> ```
> df['new_column'] = [1, 2, 3, ..., 42]
> ```
>
> It can also be set as a computation of a property from another row
>
> ```
> df['new_column'] = df['current_column'] / 4
> ```

```
df['protein1_healthy_molar_conc'] = df['A280_protein1_healthy'] / (protein_1_ext_coeff * l)
df['protein1_patient_molar_conc'] = df['A280_protein1_patient'] / (protein_1_ext_coeff * l)
df['protein2_healthy_molar_conc'] = df['A280_protein2_healthy'] / (protein_2_ext_coeff * l)
df['protein2_patient_molar_conc'] = df['A280_protein2_patient'] / (protein_2_ext_coeff * l)
display(df)
```

	A280_protein1_healthy	A280_protein1_patient	A280_protein2_healthy	A280_protein2_patient	protein1_healthy_molar_conc	protein1_patient_molar_conc	protein2_healthy_molar_conc	protein2_patient_molar_conc
0	0.086241	0.014437	0.035768	0.034756	0.000005	8.020679e-07	0.000004	0.000004
1	0.095333	0.014813	0.040662	0.040242	0.000005	8.229321e-07	0.000005	0.000005
2	0.091058	0.014119	0.039344	0.032946	0.000005	7.844136e-07	0.000005	0.000004
3	0.105004	0.014430	0.041232	0.032568	0.000006	8.016667e-07	0.000005	0.000004
4	0.097421	0.014083	0.035860	0.036829	0.000005	7.824074e-07	0.000004	0.000005
...	...	...	...	...	...	...	...	...
495	0.090870	0.015723	0.031065	0.031464	0.000005	8.734877e-07	0.000004	0.000004
496	0.084912	0.014091	0.036573	0.037934	0.000005	7.828086e-07	0.000005	0.000005
497	0.097572	0.014300	0.043926	0.046982	0.000005	7.944444e-07	0.000005	0.000006
498	0.096186	0.014589	0.040299	0.036045	0.000005	8.104938e-07	0.000005	0.000005
499	0.079806	0.013657	0.030360	0.041318	0.000004	7.587346e-07	0.000004	0.000005

(c). Concentrations in mg/mL

Add another set of four columns containing the concentrations in mg/mL.

```
protein_1_mw = 130 * 10**3
protein_2_mw = 57 * 10**3
```

```
df['protein1_healthy_conc'] = df['protein1_healthy_molar_conc'] * protein_1_mw
df['protein1_patient_conc'] = df['protein1_patient_molar_conc'] * protein_1_mw
df['protein2_healthy_conc'] = df['protein2_healthy_molar_conc'] * protein_2_mw
df['protein2_patient_conc'] = df['protein2_patient_molar_conc'] * protein_2_mw

names = ['protein1_healthy_conc', 'protein1_patient_conc', 'protein2_healthy_conc',
'protein2_patient_conc']
display(df[names])
```

	protein1_healthy_conc	protein1_patient_conc	protein2_healthy_conc	protein2_patient_conc
0	0.622848	0.104269	0.254843	0.247635
1	0.688519	0.106981	0.289719	0.286724
2	0.657640	0.101974	0.280328	0.234740
3	0.758361	0.104217	0.293781	0.232050
4	0.703593	0.101713	0.255503	0.262408
…	…	…	…	…
495	0.656283	0.113553	0.221338	0.224181
496	0.613251	0.101765	0.260580	0.270276
497	0.704688	0.103278	0.312970	0.334749
498	0.694673	0.105364	0.287130	0.256823
499	0.576373	0.098635	0.216312	0.294390

(d) Mean concentration

Now that we have the mass concentrations, calculate the mean mass concentration in the four categories.

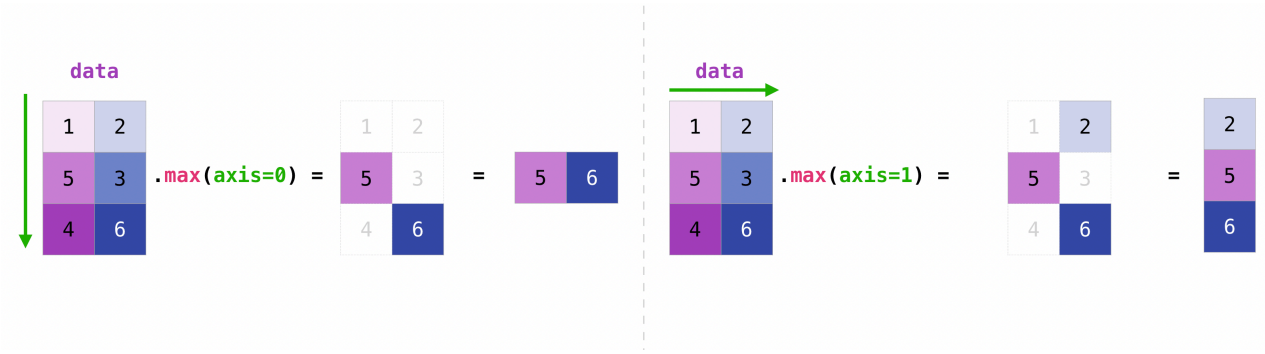
Tip

When displaying the dataframe above we indexed it with `names` as `df[names]`. We can do the same to compute something over just the four rows.

For example the if we have a `DataFrame` called `example_df`, we can calculate the mean over the **rows** as:

```
example_df[names].mean(axis=0)
```

Here `axis=0` means that we apply the operation over the first axis which by convention are the rows. The figure below visualizes this




```
mean = df[names].mean(axis=0)
unit_mean = "mg/mL"
print(mean)
print("unit_mean")
```

```
protein1_healthy_conc    0.677316
protein1_patient_conc    0.104155
protein2_healthy_conc    0.242293
protein2_patient_conc    0.256983
dtype: float64
unit_mean
```

(e) Standard deviation

Calculate the standard deviation

Tip

The standard deviation can be calculated using the `.std`-method that works in the same way as the `.mean`-method we used above.

```
std = df[names].std(axis=0)
unit_std = "mg/mL"
```

```
print(std)
print("unit_std")
```

```
protein1_healthy_conc    0.107141
protein1_patient_conc    0.005458
protein2_healthy_conc    0.040127
protein2_patient_conc    0.052373
dtype: float64
unit_std
```

(f) Analyze the results

Consider the following questions

- By comparing healthy individuals with patients, could any of the two proteins would be involved in disease development or be used as a biomarker for diagnosing the disease?

```
"""The concentration of protein1 is much lower in patients compared to the healthy
individuals, whereas protein2 is basically the same. This suggests that the loss of
protein1 was some how involved in the disease, although you cannot conclude causality
from a correlation. However, you might be able to use it as a biomarker to diagnose
whether an individual had the disease. In this case, you only need correlation,
not causation. Examples of this would include prostate specific-antigen (PSA),
which is routinely used to screen for prostate cancer."""
```

```
'The concentration of protein1 is much lower in patients compared to the
healthy\nindividuals, whereas protein2 is basically the same. This suggests that the loss
of\nprotein1 was some how involved in the disease, although you cannot conclude causality
\nfrom a correlation. However, you might be able to use it as a biomarker to
diagnose\nwhether an individual had the disease. In this case, you only need correlation,
\nnot causation. Examples of this would include prostate specific-antigen (PSA), \nwhich is
routinely used to screen for prostate cancer.'
```

- What additional information does the standard deviation provide in addition to the average value of the protein concentration?

```
"""The standard deviations provide information about the natural variation in protein
levels. This will be important for deciding whether a difference between the two
groups compared is reliable. For a larger natural variation, tyou will need a
larger difference between the groups or a a larger sample size (number of patients tested)
to conclude a difference is significant. You will return to this in a more formal
setting in statistics classes later in your studies."""
```

```
'The standard deviations provide information about the natural variation in protein
\nlevels. This will be important for deciding whether a difference between the two \ngroups
compared is reliable. For a larger natural variation, tyou will need a \nlarger difference
between the groups or a a larger sample size (number of patients tested)\nto conclude a
```

```
difference is significant. You will return to this in a more formal \nsetting in statistics
classes later in your studies.'
```

---

5 Extinction coefficient of human myoglobin

The protein of human myoglobin is given below

```
sequence = """GLSDGEWQLVLNVWGKVEADIPGHGQEVLIRLFKGHPETLEKFDKFKHLKSEDEMKASEDLKKHGA
TVLTALGGILKKKGHHEAEIKPLAQSHATKHKIPVKYLEFISECIIQVLQSKHPGDFGADAQGAMNKALELFRKDMASNY
KELGFQG"""
```

We want to calculate the *extinction coefficient* of this protein, we have seen that this can be calculated using the formula

$$\epsilon(280\text{nm}) = N_{Trp}\epsilon_{Trp} + N_{Tyr}\epsilon_{Tyr} + N_{Cys}\epsilon_{Cys} \tag{1}$$

Where N_{Trp} is the number of Tryptophan in the protein (and likewise for the other two terms), and the three constants A, B and C are given as

$$\epsilon_{Trp} = 5500\ \text{M}^{-1}\text{cm}^{-1}$$
$$\epsilon_{Tyr} = 1490\ \text{M}^{-1}\text{cm}^{-1}$$
$$\epsilon_{Cys} = 125\ \text{M}^{-1}\text{cm}^{-1}$$

In order to calculate the formula we need to know the count of the relevant residues, we can use Python to get that - for example we can count the number of Tryptophan like so;

```
N_trp = sequence.count("W")
```

(a) Determine the number of residues

In the cell below find the number of residues

```
N_tyr = sequence.count("Y")
N_cys = sequence.count("C")
```

You can check what Python has stored each variable by using `print`

```
print(N_trp)
print(f"{N_tyr = }") # This is just a way of make a string that looks nice.
print(f"{N_cys = }")
```

```
2
N_tyr = 2
N_cys = 1
```

(b) Calculate the extinction coefficient

Use equation (Equation 1) to calculate the extinction coefficient of human myoglobin.

```
eps_trp = 5500
eps_tyr = 1490
eps_cys = 125
```

```
epsilon = eps_trp * N_trp + eps_tyr * N_tyr + eps_cys*N_cys
unit_epsilon = "\mathrm{M^{-1} cm^{-1}}"
print(f"{epsilon = :.0f} {unit_epsilon}")
```

```
epsilon = 14105 \mathrm{M^{-1} cm^{-1}}
```

What are the units of this value?

(c) Compare residue count to ProtParam

ProtParam is an online tool that calculates various physical and chemical parameters from a given protein sequence and is used worldwide in research laboratories.

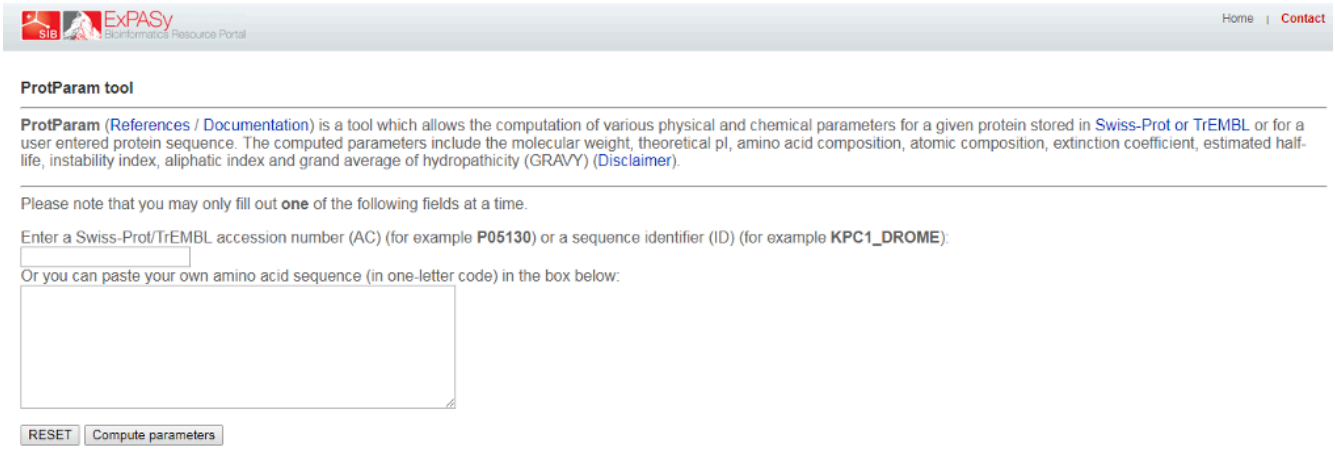


Figure 2: Protpram tool input page.

Go to ProtPram at this link: https://web.expasy.org/protparam/ and paste the sequence and click **Compute Parameters**. You should then see the calculated parameters, similar to in the image below

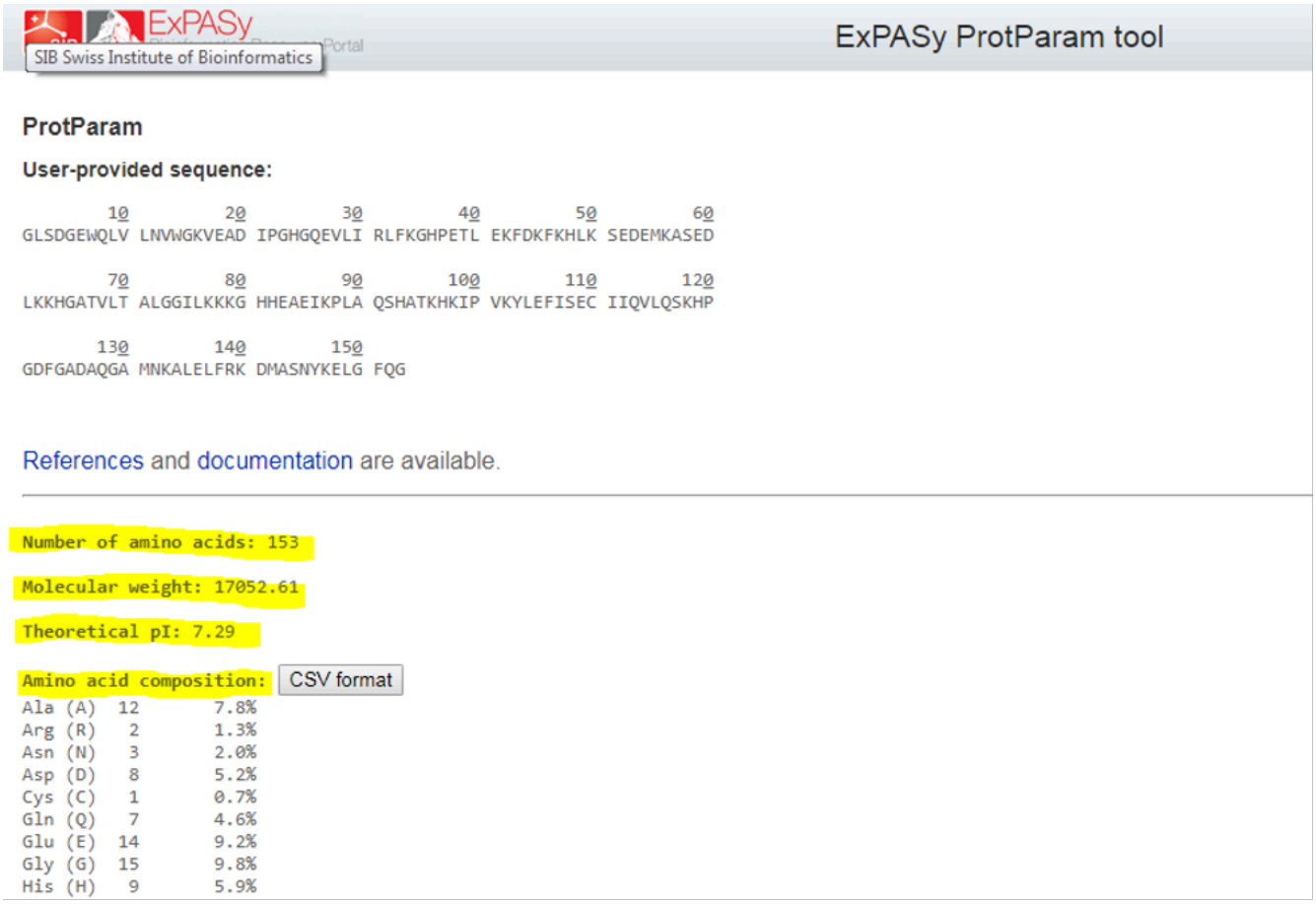


Figure 3: Protpram tool calculated properties.

On the output page you will see a calculated extinction coefficient. Does that match your calculation? If not, what is the different between the assumptions made?

(d) Calculate the absorbance

Using the extinction coefficient and the molecular weight given by ProtParam, calculate the absorbance at 280 nm of a myoglobin solution at a concentration of 1 mg/mL in a cuvette with a light path of 1 cm.

```
molecular_weight = 17052.61
path_length = 1 # cm
concentration = 1 # mg/mL
```

Remember to convert the concentration to mol/L.

```
A280 = concentration / molecular_weight * path_length * epsilon
print(A280)
```

```
0.827146108425631
```

This value is what is known as the A280(0.1%) of a protein, i.e. the absorbance of a given protein at a concentration of 0.1% weight/volume (= 1 g/L = 1 mg/mL).

(e) Titin

We have now calculated the extinction coefficient of a protein, now we will make our code more reusable so that it can be applied to other proteins easily.

> **i** Note
>
> A function in Python is a set of instructions, like a recipe, that can be defined and reused multiple times. The syntax is like this
>
> ```
> def my_function(input_1, input_2):
>     # Some operation
>     some_internal_variable = input_1 + input_2
>     ...
>     return something_weve_calculated
> ```
>
> **Line 1**
>
> The `def` command is used to define the functions name, here `my_function`, and state its inputs, e.g. the name and ingredients of a recipe.
>
> **Line 3**
>
> The body of the function defines what it does, the body can have any number of lines of code. Similar to the instructions of a recipe.
>
> **Line 5**
>
> Finally, the function can `return` something, like the final product of a recipe.
>
> Note that the function is not executed by doing this, like how a cake isn't baked by writing down the recipe, in order actually use the function it needs to be *called*
>
> ```
> output = my_function(1, 2)
> ```
>
> This is also how we have already used other functions like `print`.

The way of doing so is by defining a function that does the necessary operations for a given sequence. In this way the code can be reused for any sequence.

Finish implementing the body of the function below, note that you have already written all the required code - you just need to copy it into the function.

```
def extinction_coefficient(sequence):

    # Start by counting:
    N_trp = sequence.count("W")
    N_tyr = sequence.count("Y")
    N_cys = sequence.count("C")

    # Define the residue extinction coeffiecients
    eps_trp = 5500
```

```
    eps_tyr = 1490
    eps_cys = 125

    # Calculate
    epsilon = eps_trp * N_trp + eps_tyr * N_tyr + eps_cys*N_cys
    unit_epsilon = "\mathrm{M^{-1} cm^{-1}}"
    print(f"{epsilon = :.0f} {unit_epsilon}")
```

It's always a good idea to check that functions do what we expect, so we can confirm that it gives the same result for human myoglobin as we calculated before

```
epsilon_func = extinction_coefficient(sequence)
print(f"Epsilon: {epsilon}")
print(f"Epsilon from function: {epsilon_func}")
```

```
epsilon = 14105 \mathrm{M^{-1} cm^{-1}}
Epsilon: 14105
Epsilon from function: None
```

The largest known protein is Titin, the cell below loads the sequence of titin and prints a few bits of information about it. (You can also find the full sequence in the dataset `extin-coeff-human-myogl.txt`.)

```
from fysisk_biokemi.datasets import load_dataset
titin_sequence = load_dataset('titin')

print(f'Titin number of residues: {len(titin_sequence)}')
print('First 100 residues')
print(titin_sequence[0:100])
```

```
Titin number of residues: 34922
First 100 residues
MTTQAPTFTQPLQSVVVLEGSTATFEAHISGFPVPEVSWFRDGQVISTSTLPGVQISFSD
GRAKLTIPAVTKANSGRYSLKATNGSGQATSTAELLVKA
```

Use your function to calculate the extinction coefficient of titin.

```
titin_eps = extinction_coefficient(titin_sequence)
print(titin_eps)
```

```
epsilon = 4115635 \mathrm{M^{-1} cm^{-1}}
None
```

Tip

You don't need to understand the code below, it's just ment to illustrate that knowing some Python will allow you to explore the topics that interest you in more detail.

In general Python is very powerful at letting us explore properties of sequences, for example the cell below calculates number of residues between each Tryptophan in the Titin sequence and plot the distribution.

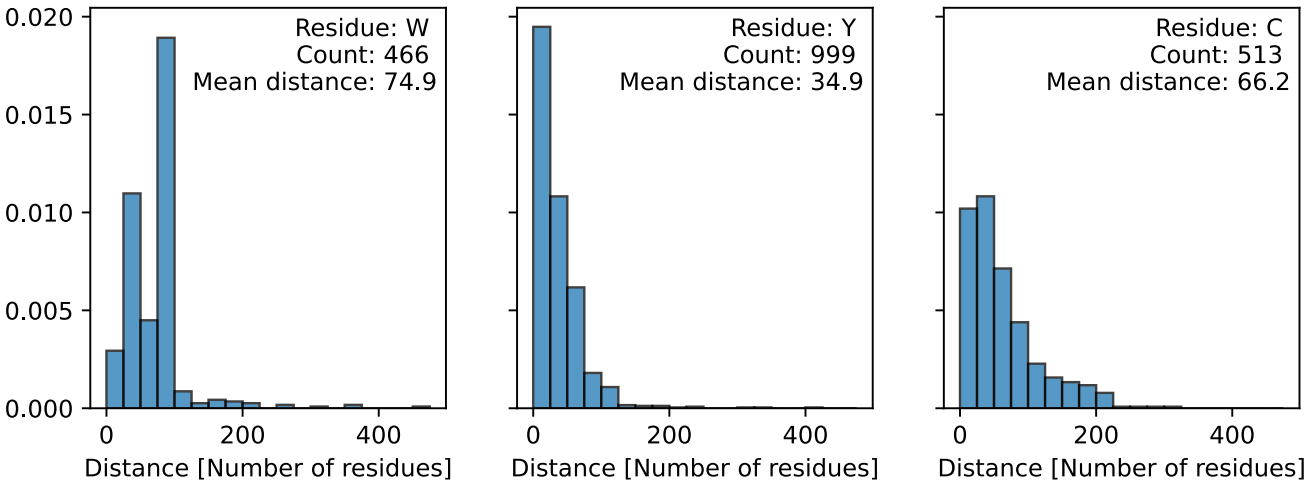
```
import numpy as np
import matplotlib.pyplot as plt

def get_distance(sequence, letter):
    W_index = np.argwhere(np.array([l for l in sequence]) == letter)
    count = len(W_index)
    distance = (W_index - np.roll(W_index, 1))[1:]
    return distance, count

letters = ['W', 'Y', 'C']

fig, axes = plt.subplots(1, 3, figsize=(3*3, 3), sharey=True)
axes = axes.flatten()

for ax, letter in zip(axes, letters):
    distance, count = get_distance(titin_sequence, letter)
    ax.hist(distance, bins=np.arange(0, 500, 25),
        edgecolor='black', alpha=0.75, density=True)
    ax.set_xlabel('Distance [Number of residues]')
    info = f'Residue: {letter} \nCount: {count} \nMean distance: {np.mean(distance):.1f}'
    ax.text(0.975, 0.975, info, transform=ax.transAxes, ha='right', va='top')
```



6 Tryptophan absorption and fluorescence

```
from fysisk_biokemi.datasets import load_dataset
from IPython.display import display
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
pd.set_option('display.max_rows', 6)
```

You investigate a protein that neither contains tyrosine nor cysteine residues. A 37 µM solution gives an absorbance of 0.41 at 280nm at a light path of 1 cm.

(a) Number of residues

How many tryptophan residues does the protein contain?

(b) Concentration

You would like to conduct a protein stability experiment at an absorbance of 0,8 (path length 1cm). What concentration should you use?

(c) Plot the spectra

The cell below loads extinction and emission spectra for tryptophan in aqueous buffer. (The data files are `trypt-absor-fluor-emission.xlsx` and `trypt-absor-fluor-extinction.xlsx`)

```
df_emission = load_dataset('week47_1_emi')
df_extinction = load_dataset('week47_1_ext')
display(df_emission)
display(df_extinction)
```

	wavelength_(nm)	emission_(AU)
0	280.0	379
1	280.5	396
2	281.0	403
…	…	…
396	478.0	5024
397	478.5	4950
398	479.0	4813

	wavelength_(nm)	molar_extinction_(cm-1/M)
0	219.75	12488
1	220.00	12872
2	220.25	12428
…	…	…
396	318.75	14

	wavelength_(nm)	molar_extinction_(cm-1/M)
397	319.00	18
398	319.25	45

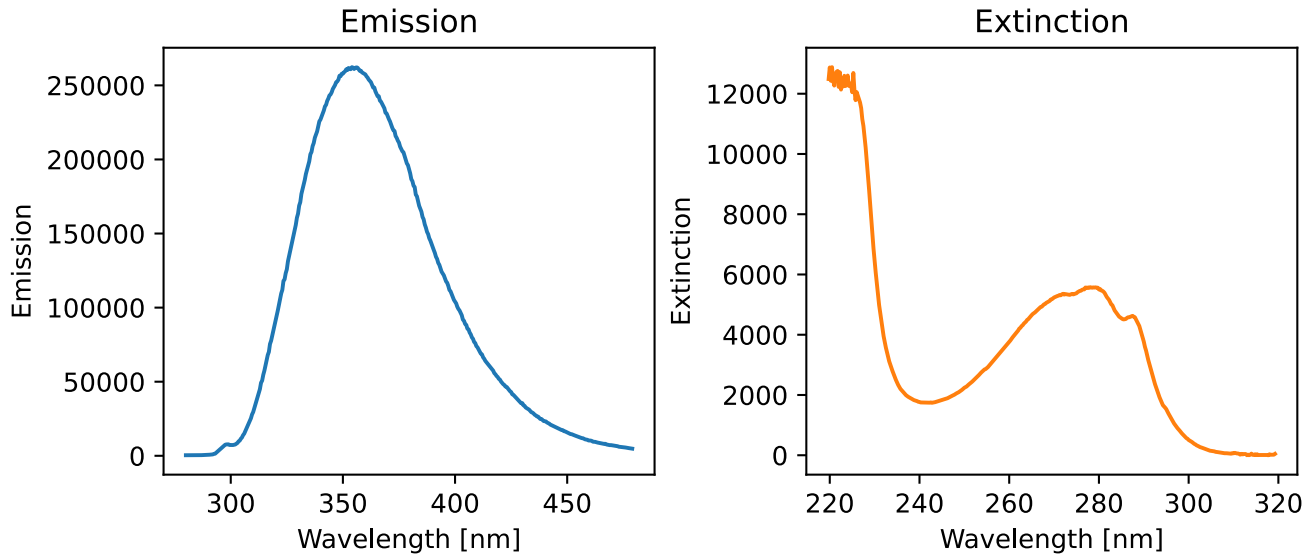
Make plot showing the two spectra

```
fig, axes = plt.subplots(1, 2, figsize=(7, 3), layout='constrained')

# Emission
ax = axes[0]
ax.set_title('Emission')
ax.plot(df_emission['wavelength_(nm)'], df_emission['emission_(AU)'], label='Emission',
color='C0')
ax.set_ylabel('Emission')
ax.set_xlabel('Wavelength [nm]')

# Extinction
ax = axes[1]
ax.set_title('Extinction')
ax.plot(df_extinction['wavelength_(nm)'], df_extinction['molar_extinction_(cm-1/M)'],
label='Extinction', color='C1')

# Customization
ax.set_ylabel('Extinction')
ax.set_xlabel('Wavelength [nm]')
plt.show()
```



(d) Stokes shift

From the plot estimate the Stokes shift

```
estimated_stokes_shift = 30
unit_stokes_shift = "nm"
print(f"{estimated_stokes_shift = :.0f} {unit_stokes_shift}")
```

```
estimated_stokes_shift = 30 nm
```

Compare your estimated Stokes shift to the output of the cell below that computes the stokes shift.

```
max_index = np.argmax(df_emission['emission_(AU)'])
emission_wavelength_max = df_emission['wavelength_(nm)'][max_index]

max_index = np.argmax(df_extinction['molar_extinction_(cm-1/M)'])
extinction_wavelength_max = df_extinction['wavelength_(nm)'][max_index]

stokes_shift = emission_wavelength_max - extinction_wavelength_max

print(f"{emission_wavelength_max = :.1f} nm")
print(f"{extinction_wavelength_max = :.1f} nm")
print(f"{stokes_shift = :.1f} nm")
print(f"{estimated_stokes_shift = :.1f} nm")
```

```
emission_wavelength_max = 354.0 nm
extinction_wavelength_max = 220.5 nm
stokes_shift = 133.5 nm
estimated_stokes_shift = 30.0 nm
```

(e) pH and how it changes properties.

Next, you compare the fluorescence emission spectra for your protein at pH 7 and at pH 2. You can find the data in the files `trypt-absor-fluor-ph2.xlsx` and `trypt-absor-fluor-ph7.xlsx` - the cell below also loads these datasets.

The datasets are loaded in a different way than you've seen before - as help to you as this exercise deals with multiple datasets. The datasets could also be loaded with the widget you've seen before

```
df_ph2 = load_dataset('week47_1_ph2')
df_ph7 = load_dataset('week47_1_ph7')
display(df_ph2, df_ph2)
```

	Wavelength(nm)	Fluo_Int
0	311.623037	0.890000
1	314.450262	1.128668
2	315.863874	1.128668
...	...	...
43	367.068063	27.990971

	Wavelength(nm)	Fluo_Int
44	368.638743	27.313770
45	370.052356	26.636569

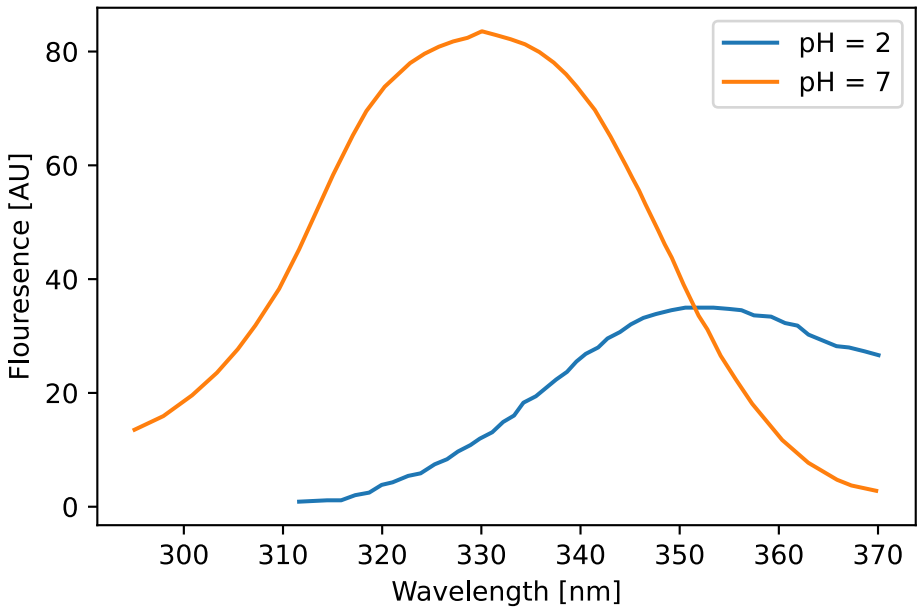
	Wavelength(nm)	Fluo_Int
0	311.623037	0.890000
1	314.450262	1.128668
2	315.863874	1.128668
...	...	...
43	367.068063	27.990971
44	368.638743	27.313770
45	370.052356	26.636569

Make a plot comparing the two emission spectra. Which resembles the spectrum of tryptophan in water most?

```
fig, ax = plt.subplots()

ax.plot(df_ph2['Wavelength(nm)'], df_ph2['Fluo_Int'], label='pH = 2')
ax.plot(df_ph7['Wavelength(nm)'], df_ph7['Fluo_Int'], label='pH = 7')

ax.set_xlabel('Wavelength [nm]')
ax.set_ylabel('Flouresence [AU]')
ax.legend()
plt.show()
```



(f) Explanation.

Explain why the same protein gives such different spectra at pH 2 and pH 7.

```
"""
The protein unfolds at acidic pH. Therefore the Trp residues are exposed to water
at pH 2, but buried in the hydrophobic core (and thus blue-shifted) at pH 7
"""
```

---

7 The fluorescent protein “mCherry”.

```
import pandas as pd
from IPython.display import display
```

The spectra of many fluorescent proteins can be found at the website: www.fpbase.org. Go to FPbase and search for “mCherry”.

(a) Find parameters.

Find the following parameters for the protein

- Extinction coefficient at absorbance maximum
- Quantum yield
- The organism from which it was originally isolated
- Molecular weight

Save them to seperate variables in the cell below.

```
QY = 0.22
EC = 72000 # [1/(M cm)]
organism = "Discosoma sp."
m_w = 26.7 * 10**3 # From fpbase
```

(b) Absorbance

What is the absorbance of a 1 µM solution of mCherry at its absorption maximum at a path length of 1 cm?

```
c = 1 * 10**(-6) # [M]
l = 1 # [cm]
A_max = EC * c * l
print(f"{A_max = :3.3f}")
```

```
A_max = 0.072
```

(c) Extinction coefficient from sequence.

> **Note**
>
> We will treat the sequence as a `str`, like any other text, `str`'s are defined like
>
> ```
> one_line_string = "This is some text, that makes up my string."
> ```
>
> Which uses "-quotation marks around the text, for longer strings it can be useful to instead use
>
> ```
> multi_line_string = """This is a very long text, so long in fact that it
> takes up multiple lines and I therefore use a slightly different syntax.
> To make the string longer I will confess that I am hungry right now.
> """
> ```

The sequence of the protein is also given. From this determine the extinction coefficient at 280 nm.

Start by taking the sequence from the website and assigning it to the variable `sequence` in the cell below.

```
sequence="""MVSKGEEDNM AIIKEFMRFK VHMEGSVNGH EFEIEGEGEG RPYEGTQTAK LKVTKGGPLP FAWDILSPQF
MYGSKAYVKH PADIPDYLKL SFPEGFKWER VMNFEDGGVV TVTQDSSLQD GEFIYKVKLR GTNFPSDGPV MQKKTMGWEA
SSERMYPEDG ALKGEIKQRL KLKDGGHYDA EVKTTYKAKK PVQLPGAYNV NIKLDITSHN EDYTIVEQYE RAEGRHSTGG
MDELYK"""
```

Now use the sequence to calculate the extinction coefficient, finish the code below (or take the function you implemented in a previous exercise!)

```
# This is 'dictionary' with the extinction coefficients of the relevant
# amino acid residues. Dictionaries are indexed with 'keys', so you can retrieve
# the value for W as: ext_residue["W"].
ext_residue = {"W": 5500, "Y": 1490, "C": 125}

# Count the number of active residues
N_trp = sequence.count("W")
N_tyr = sequence.count("Y")
N_cys = sequence.count("C")

# Calculate the extinction coefficient:
ext_coeff = N_trp * ext_residue["W"] + N_tyr * ext_residue["Y"] + N_cys * ext_residue["C"]
unit_epsilon = "\mathrm{M^{-1} cm^{-1}}"
print(f"{ext_coeff = :3.3f} {unit_epsilon}")
```

```
ext_coeff = 34380.000 \mathrm{M^{-1} cm^{-1}}
```

(d) Concentration from A280nm

What is the molar concentration of solution of mCherry with an A_280nm of 0.45

```
A280 = 0.45
c_mol = A280 / (ext_coeff)
unit_c_mol = "M"
print(f"{c_mol = :3.3f} {unit_c_mol}")
```

```
c_mol = 0.000 M
```

The excitation and emission spectra can be downloaded as a csv-file by clicking the download icon as highlighted below

Figure 4: Screenshot of the download button on fpbase

(e) Load the dataset

Go to www.fpbase.org and download the spectrum as described above.

Load the dataset as a `DataFrame` with pandas. Set `data_file_path` to the path of the downloaded dataset file.

```
from IPython.display import display
from fysisk_biokemi.datasets import load_dataset
df = load_dataset('mCherry') # Load from package for the solution so it doesn't require to interact.
display(df)
```

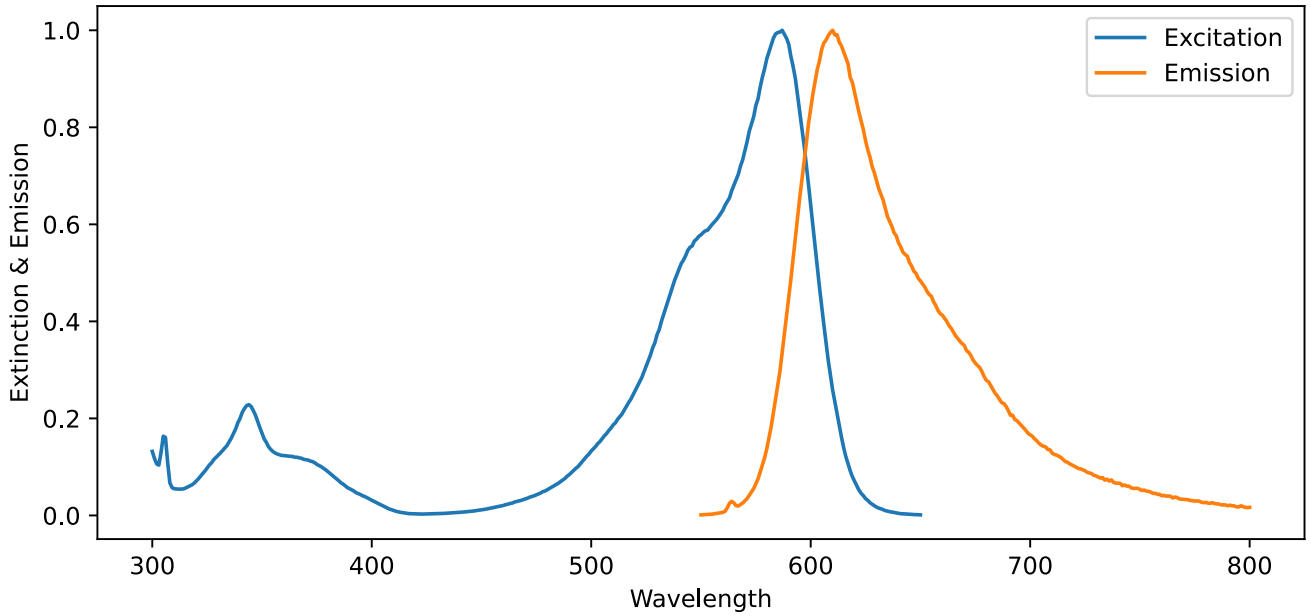
	wavelength	mCherry ex	mCherry em	mCherry 2p
0	300	0.1320	NaN	NaN
1	301	0.1175	NaN	NaN
2	302	0.1063	NaN	NaN
…	…	…	…	…
1097	1397	NaN	NaN	0.0
1098	1398	NaN	NaN	0.0
1099	1399	NaN	NaN	0.0

(f) Plot spectra

Make your own plot showing the excitation and emission spectra of “mCherry” using the above data.

```
import matplotlib.pyplot as plt
fig, ax = plt.subplots(figsize=(9, 4))
ax.plot(df['wavelength'], df['mCherry ex'], label='Excitation')
ax.plot(df['wavelength'], df['mCherry em'], label='Emission')
ax.legend()
ax.set_xlabel('Wavelength')
```

```
ax.set_ylabel('Extinction & Emission')
plt.show()
```



(g) Stokes shift

Estimate the Stokes shift of mCherry from the plot.

```
estimated_stokes_shift = 30 # Your estimation
unit_stokes_shift = "nm"
```

The cell below shows how this could calculated using Python

```
import numpy as np
# Start by extracting the wavelengths from the DataFrame
wavelengths = df['wavelength']

# Find the indices
ex_max_idx = np.argmax(df['mCherry ex'])
em_max_idx = np.argmax(df['mCherry em'])

# Use the index to find the corresponding wavelengths
lambda_ex = wavelengths[ex_max_idx]
lambda_em = wavelengths[em_max_idx]

stokes_shift = lambda_em - lambda_ex

print(f"{lambda_ex = :d}")
print(f"{lambda_em = :d}")
print(f"{stokes_shift = :d}")
print(f"{estimated_stokes_shift = :d}")
```

```
lambda_ex = 587
lambda_em = 610
stokes_shift = 23
estimated_stokes_shift = 30
```

Tip

If you have two arrays `A` and `B` you can find the entry in `A` corresponding to the largest value in `B` like this

```
A_at_B_max = A[np.argmax(B)]
```

The `np.argmax` stands for **arg**ument **max**imum meaning that it finds the index of the maximum value in a given array. The figure below illustrates this

Figure 5: Illustration of argmax

(h) Colors

Compare to a plot of the visual spectrum. What colors are the light that corresponds to the excitation and emission maxima respectively?

Why do you think the protein is called *Cherry*?

---

8 Python: Introduction to curve fitting

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.optimize import curve_fit
```

Curve fitting is a fundamental skill in biochemistry and biophysics for analyzing experimental data. We use curve fitting to determine parameters in mathematical models that describe biological processes like enzyme kinetics, binding affinity, and reaction rates.

The basic idea is to find the parameters of a mathematical function that best describes our experimental data.

(a) Understanding the problem

Real biochemical data is often requires complex fitting functions. Therefore, we will start out with a simpler data that follows a linear relationship. The data below represents a theoretical experiment where we measure some output y at different input values x:

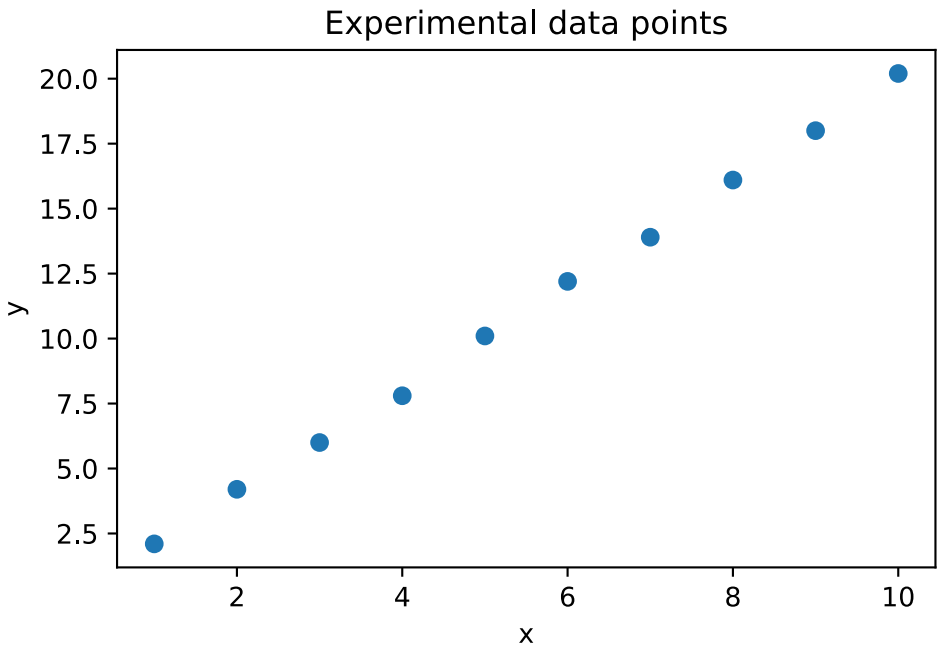
```
## This makes two arrays (columns) ##
## One with the x coordinates ##
## One with the y coordinates ##
x_data = np.array([1, 2, 3, 4, 5, 6, 7, 8, 9, 10])
y_data = np.array([2.1, 4.2, 6.0, 7.8, 10.1, 12.2, 13.9, 16.1, 18.0, 20.2])
```

The functions above simply make two arrays - i.e. columns of data - containing the number. You can view the content of these colmns by with `print(x_data)` or `print(y_data)`.

```
# Plot the data
fig, ax = plt.subplots()

## Your task: Plot the data as a scatter plot! ##
## Recall that you can do this with: ax.plot(x_data, y_data, 'o')
ax.plot(x_data, y_data, 'o')

## EXTRAS - You do not need to read or understand ##
## Adding customization ##
ax.set_xlabel('x')
ax.set_ylabel('y')
ax.set_title('Experimental data points');
```



Looking at this data, we can see it roughly follows a straight line. Let's assume our model is:

$$y = ax + b$$

where a and b are parameters we want to determine from the data.

(b) Defining a fitting function

To use `scipy.optimize.curve_fit`, we need to define a function where:

- The **first** argument is the independent variable (x)
- The remaining arguments are the parameters to be fitted

Define a linear function for fitting:

```
def linear_function(x, a, b):
    ## Your task: Calculate the linear function ##
    result = a * x + b
    return result
```

Always a good idea to check that your function works as expected

As a test we are adding 1 in the place of x, 2 in the place of a and 0 in the place of b. In other words: $1 \cdot 2 + 0 = 2$

```
linear_function(1, 2, 0) # Should give 2
```

When in doubt it is a good idea to do sanity checks like this.

(c) Your first curve fit

Now we can use `curve_fit` to find the best parameters. The basic syntax is:

```
fitted_parameters, trash = curve_fit(function, x_data, y_data, p0=initial_guess)
```

Where:

- `function`: The function we defined above
- `x_data`, `y_data`: Our experimental data
- `p0`: Initial guess for parameters (optional but recommended)
- `fitted_parameters`: The optimized parameters (what we want!)
- `trash`: Don't worry about this.

Finish the cell below to perform the curve fit by adding the three missing arguments to the call to `curve_fit`.

```
# Initial guess: a=2, b=0
initial_guess = [2, 0]

# Perform the fit
fitted_parameters, trash = curve_fit(linear_function, x_data, y_data, p0=initial_guess)

# Extract parameters
a_fit, b_fit = fitted_parameters

print(fitted_parameters)
print(a_fit)
print(b_fit)
```

```
[2.00242424 0.04666667]
2.0024242424242424
0.04666666666666669
```

(d) Visualizing the fit

It's crucial to always plot your fit to see how well it describes the data. To do this we evaluate the function with the fitted parameters and a densely sampled independent variable.

```
x_smooth = np.linspace(0, 11, 100) # Makes 100 equally spaced points between 0 and 11.
y_fit = linear_function(x_smooth, a_fit, b_fit) # Call the linear function using x_smooth,
a_fit and b_fit
```

Then we can plot it

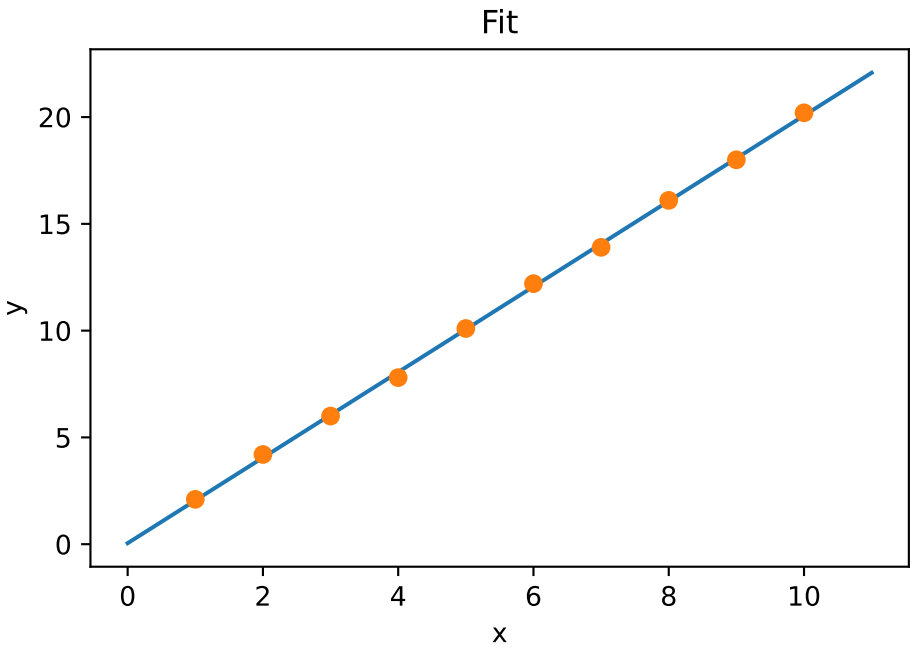
```
fig, ax = plt.subplots()

## Your task: Plot the fitted function using x_smooth and y_fit ##
ax.plot(x_smooth, y_fit)

## Plots the experimental data ##
ax.plot(x_data, y_data, 'o')

## EXTRA ##
## You do not need to understand this ##
## Customizes the plot ##
ax.set_xlabel('x')
ax.set_ylabel('y')
ax.set_title('Fit')
```

```
Text(0.5, 1.0, 'Fit')
```



(e) Nonlinear fitting: Exponential decay

Many biological processes follow nonlinear relationships. Let's work with exponential decay, which is common in biochemistry (e.g., radioactive decay or unimolecular chemical reactions).

You can load the dataset from the `exp_decay_data.xlsx`-file with the widget below

```
from fysisk_biokemi.widgets import DataUploader
from IPython.display import display
uploader = DataUploader()
uploader.display()
```

Run the next cell **after** uploading the file

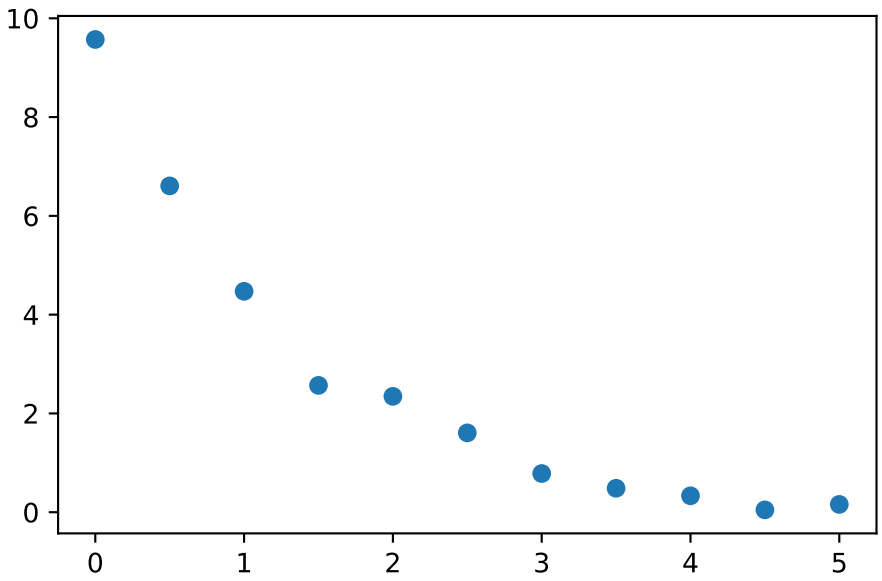
```
df = uploader.get_dataframe()
display(df)
```

```
from IPython.display import display
from fysisk_biokemi.datasets import load_dataset
df = load_dataset('exp_decay_data')
display(df)
```

	time	signal
0	0.0	9.571655
1	0.5	6.606815
2	1.0	4.472434
…	…	…
8	4.0	0.332479
9	4.5	0.047604
10	5.0	0.158024

Now make a plot of the data

```
## Your task: Plot the data from the dataframe ´df´. ##
## Recall that your get the time-column as df['time'] ##
## Similarly you get the signal-column  as df['signal'] ##
fig, ax = plt.subplots()
ax.plot(df['time'], df['signal'], 'o')
```



The model we want to fit is:

$$\text{signal} = Ae^{-kt}$$

Define the exponential decay function:

```
def exponential_decay(t, A, k):
    ## Your task: Calculate the exponential function ##
    ## Remember that np.exp calculates the exponential function ##
    result = A * np.exp(-k * t)
    return result

## This calculates the function for t=1, A=1, k=1 ##
## If you did it correctly you should see 0.3678... ##
test_result = exponential_decay(1, 1, 1)
print(test_result)
```

```
0.36787944117144233
```

Now fit the exponential function to the data:

```
# Initial guess: A=8, k=1
initial_guess = [8, 1]

# Perform the fit
fitted_parameters, trash = curve_fit(exponential_decay, df['time'], df['signal'],
p0=initial_guess)

# Extract parameters
A_fit, k_fit = fitted_parameters
```

```
print(A_fit)
print(k_fit)
```

```
9.627936471422466
0.7845023305388426
```

Again we should plot to check that it looks as expected

```
## This makes 100 linearly spaced points between 0 and 5 ##
t_smooth = np.linspace(0, 5, 100)

## Your task: Calculate the function using the fitted parameters and t_smooth ##
signal_fit = exponential_decay(t_smooth, A_fit, k_fit)
```

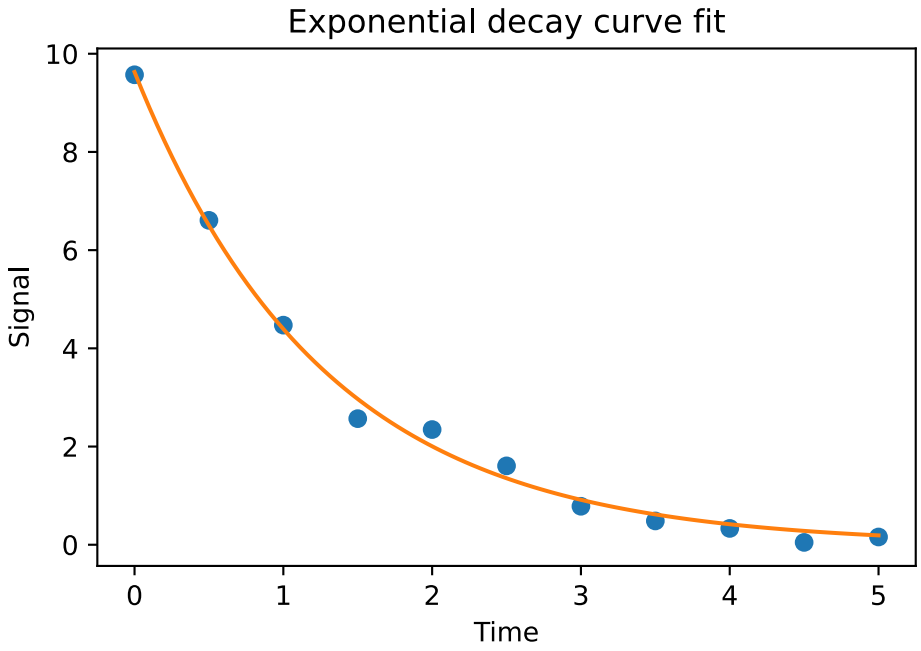
Now we can plot the fit along with the data.

```
fig, ax = plt.subplots()

## Your task: Plot the observations - you can copy this line from the ##
## previous plot. ##
ax.plot(df['time'], df['signal'], 'o')

## Your task: Plot the fit using t_smooth and signal_fit
ax.plot(t_smooth, signal_fit)

## EXTRA - You do not need to understand this ##
## This adds customization ##
ax.set_xlabel('Time')
ax.set_ylabel('Signal')
ax.set_title('Exponential decay curve fit')
plt.show()
```



Key points for successful curve fitting:

1. **Always plot your data first** to understand what kind of function might fit
2. **Always plot the fit** to visually check if it makes sense
3. **Check if parameters are reasonable** based on your biochemical knowledge

! Summary of `curve_fit`

Fitting refers to finding the parameters that make an assumed functional form best 'fit' the data.

In Python we will use the `curve_fit` from the `scipy` package to do so. The function looks like this

```
curve_fit(function,
            x_data,
            y_data,
            p0=[param_1, param_2, ...])
```

The arguments are

- `function`: A python function where the **first** argument is the independent variable (typically the x coordinate), and other arguments are the parameters of the functions.
- `x_data`: The observed values of the independent variable (The x coordinates).
- `y_data`: The observed values of the dependent variable (The y coordinates) .
- `p0`: Initial guesses for the parameters.

When called `curve_fit` starts by calculating how well the functions fits the data with the initial parameters in `p0` and then iteratively improves the fit by trying new values for the parameters in an intelligent way.

The found parameters will generally depend on `p0` and it is therefore necessary to provide a good (or good enough) guess for `p0`.

Generally, the best way to learn more about a function is to read it's documentation and then play around with it. The documentation is in this case on the SciPy website. You don't need to read it, unless you want more details.

You now have the fundamental skills needed to fit curves to biochemical data! In the exercises, you'll apply these techniques to analyze real experimental data and extract meaningful biological parameters.